

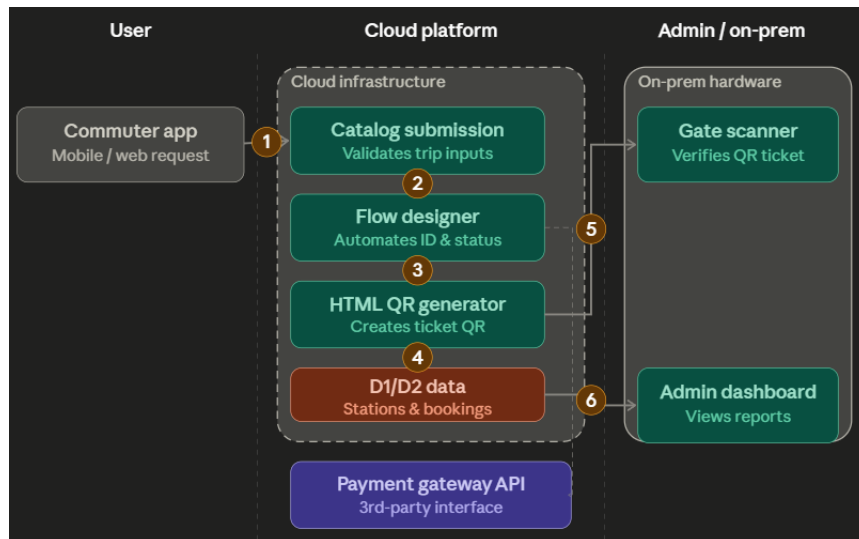
Project Design Phase-II Technology Stack (Architecture & Stack)

Date	02 August 2026
Team ID	
Project Name	Metro Ticket Booking In Servicenow
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode



Guidelines:

1. Commuter submits source, destination & ticket count
 2. Catalog submission passes validated data to flow designer
 3. Flow designer generates record ID, sends to QR generator
 4. Booking & status are written to D1/D2 data stores
 5. QR ticket is issued and read by the gate scanner
 6. Admin queries data stores for booking reports
- Dashed side line: external payment gateway API call. No ML/AI model interface in this workflow.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Logic for a process in the application	Java / Python
3.	Application Logic-2	Logic for a process in the application	IBM Watson STT service
4.	Application Logic-3	Logic for a process in the application	IBM Watson Assistant
5.	Database	Data Type, Configurations etc.	MySQL, NoSQL, etc.
6.	Cloud Database	Database Service on Cloud	IBM DB2, IBM Cloudant etc.
7.	File Storage	File storage requirements	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	Purpose of External API used in the application	IBM Weather API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source / Web Frameworks	Lightweight HTML container rendering and REST API integration	Web REST APIs, HTML/CSS
2	Security Implementations	Access control lists (ACLs) restricting ticket modification and role-based access	ServiceNow Access Control Lists (ACLs) & Role-Based Access (RBAC)
3	Scalable Architecture	Cloud-native multi-tenant SaaS architecture supporting concurrent ticket bookings	ServiceNow SaaS Cloud Platform Architecture
4	Availability	High availability with redundant cloud server infrastructure and 24/7 uptime	ServiceNow High Availability (HA) Multi-Instance Architecture
5	Performance	Optimized query execution using reference fields (Display = true) and asynchronous client scripts	ServiceNow Indexing & Asynchronous Client Scripting